

John Kangethe

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Profile

Software Engineer with 5+ years of experience building and maintaining web applications and cloud-enabled systems, including sustained open-source work on large-scale technology detection and signature maintenance. Currently pursuing an M.S. in Computer Science (Artificial Intelligence) at the University of South Dakota, focused on applied deep learning, information retrieval, and data-driven systems. Strengths include JavaScript/Python, API design, reliable deployments, and accessible, user-centered UI. Aspiring roles that blend software engineering with data/AI and research or user-facing product support.

Education

- Aug 2025 – **MSc. in Computer Science (Artificial Intelligence)**, *University of South Dakota*, Vermillion, SD
- 2023 **Certificate in Software Engineering**, *Flatiron School*, New York, USA
- 2018 – 2022 **Bachelor of Technology**, *Technical University of Kenya*, Nairobi, Kenya
- 2019 – 2021 **Diploma in Computer Science**, *Pan Africa Christian University*, Nairobi, Kenya

Skills

- CS Foundations: Data structures & algorithms, OOP, software design fundamentals, information retrieval basics, AI/ML/DL.
- Languages: JavaScript, Python, Ruby, SQL, HTML5/CSS3, Markdown.
- Frontend: Vue 3, React, Next.js, Naive UI, Vuetify, PrimeVue, Vuesax, Tailwind CSS, Streamlit
- Backend: Node.js, Express, Django, Flask, FastAPI, Ruby on Rails, REST APIs, Token-based auth, background jobs.
- Databases: PostgreSQL, MySQL, SQLite, Prisma ORM, Sequelize, schema design, migrations, query optimizations.
- AI/ML/DL: PyTorch, Pandas, NumPy, Matplotlib, scikit-learn, D3.js.
- Cloud/DevOps: Git/GitHub, Docker, Railway, Render, Vercel, Mercurial, CI/CD, Agile, Linux/CLI.
- Testing/Quality: Unit/integration testing basics (pytest/Jest); debugging, logging, code review practices.
- UX & Docs: Accessibility-minded UI, clear information hierarchy, plain-language documentation, technical writing.
- Tools: VS Code, Postman, Figma, Miro, Google Colab, Jupyter, Atom, StarUML

Professional Experience

- Jan 2023 – **Software Maintenance Engineer, BuiltWith (Australia), Remote**
Present
- Research and add technology-detection rules (regex, headers, scriptSrc), review PRs, and triage issues to keep signatures current.
 - Improve detection precision by reducing false positives/negatives, documenting edge cases, and contributing tests, fixtures, and reproducible examples.
 - Maintain and refactor signature patterns for performance (safer regex, fewer backtracking hotspots) while preserving coverage across modern stacks.
 - Debug misdetections using browser DevTools (network traces, response headers, DOM inspection), then translate findings into robust detection logic.
 - Analyze real-world sites to validate detections, compare competing signals, and tune confidence via rule ordering, exclusions, and fallback heuristics.
 - Collaborate with maintainers to align changes with coding standards, release processes, and backwards compatibility expectations.
 - Write concise contributor docs and FAQs that support community onboarding, rule authoring consistency, and faster issue resolution.
- Jan 2024 – **Cloud Solutions Practitioner, Oracle Data Center, Morocco, Africa**
July 2025
- Implemented and operated cloud-native services across dev/stage/prod, containerizing workloads with Docker and supporting repeatable deployments and environment parity.
 - Improved delivery reliability by tightening CI/CD hygiene (build/test steps, artifact/version tagging, configuration management) and standardizing release checklists.
 - Monitored service health and performance using logs/metrics/alerts; investigated incidents via root-cause analysis and implemented corrective/preventive actions.
 - Troubleshoot networking and application issues (DNS, TLS/SSL, HTTP errors, timeouts) and collaborated with engineering teams to resolve defects and reduce repeat incidents.
 - Partnered with stakeholders to scope small platform improvements (automation, tooling, workflow refinements) that reduced support turnaround time and manual effort.
- Feb 2023 – **Software Engineer, Emmerce Kenya, Nairobi, Kenya**
Dec 2023
- Built secure, scalable ecommerce features (APIs, DB tuning, third-party integrations) and optimized page performance.
 - Improved developer experience with linting, code reviews, and release notes; contributed to analytics dashboards.
- Apr 2023 – **Frontend Engineer / UI-UX Designer, ULG Africa, Nairobi, Kenya**
Feb 2024
- Delivered component-based UIs in Vue/React; emphasized accessibility, responsive patterns, and clear information hierarchy.
 - Collaborated with backend and design to iterate quickly on user feedback and product requirements.
- Jan 2023 – **Software Developer Intern, iTalanta Agency, Nairobi, Kenya**
Mar 2023
- Contributed to platform features, testing, and agile ceremonies; practiced code reviews and pair programming.
- Nov 2021 – **Informatics Intern, KEMRI, CBRD Dept., Nairobi, Kenya**
May 2022
- Supported data operations and software integrations for lab workflows; assisted with database maintenance and automation scripts.

Selected Projects

- 2025 **StudyBuddy AI (Offline Notes/PDF → Flashcards + Quiz)**
- Built an in-browser study assistant that ingests notes/PDF text and generates structured flashcards and quizzes for active recall, designed to run without a paid backend.
 - Implemented a clean Vue 3 UI for import, review, and practice modes; added progress tracking and lightweight scoring to support iterative learning.
 - Focused on privacy-first, offline-friendly workflows and export-ready study sets (JSON/CSV) for reuse across courses.

2024 **Car Rental Web App (Vue 3 + Naive UI, Node.js/Express, Prisma/PostgreSQL),**

- Built a customer booking experience and an admin dashboard with CRUD, image uploads, booking approvals, and downloadable PDF receipts/history reports.
- Designed REST APIs and data models for cars, bookings, users, and status transitions; enforced availability rules to prevent double booking.
- Implemented auth and role-based access; added admin metrics endpoints (active/pending bookings, cancellations, daily bookings, revenue estimates).

2023 **BuiltWith / Regex-Style Technology Signatures**

- Developed and maintained technology-detection signatures using regex, headers, DOM signals, and scriptSrc patterns; reduced false positives at scale.
- Implemented version extraction via capture groups and normalization; validated changes with test cases and real-site verification.

References

Available on request.